# Orange Problem Assignment

## Course: Microprocessor and Computer Architecture (UE24CS251B)

---

## 1. Introduction

This document describes the **Orange Problem Assignment** designed for the course *Microprocessor and Computer Architecture*.

Students will be required to independently learn a modern microprocessor architecture such as RISC-V and gain hands-on experience with an associated development toolchain. The assignment focuses on understanding the **RISC-V processor** architecture through hands-on assembly-level programming and simulation using **RARS (RISC-V Assembler and Runtime Simulator),** demonstrating correct use of architectural features and instructions. Evaluation will be based on the student's conceptual understanding of the chosen architecture, effective use of the tool and development environment, and the correctness, clarity, and outcomes of the implemented code, including analysis of results.

---

## 2. Tool Used: RARS (RISC-V Assembler and Runtime Simulator)

**RARS** is a lightweight and user-friendly simulator specifically designed for beginners learning RISC-V assembly language. It provides an integrated graphical environment for writing, assembling, executing, and debugging RISC-V programs.

### Key Features

- Graphical interface for easy interaction
- Supports RISC-V RV32I instruction set
- Step-by-step execution and debugging support
- Register and memory visualization
- Error highlighting and runtime exception handling

### System Requirements

- Java 8 or newer must be installed on the system
- Download the latest `rars.jar` file from the given website.

---

## 3. Assignment Description: Orange Problem

The **Orange Problem** is a RISC-V assembly-based problem where students are required to: Implement logic using RISC-V instructions, simulate the execution using RARS, observe register and memory changes and produce correct and formatted output.

The assignment consists of **two questions**

Students must submit:

- The RISC-V assembly code (`.asm` file).
- A brief explanation of the code logic.
- Output screenshot.

---

## 4. Evaluation Rubric

The assignment is evaluated for a total of **10 marks**, split as follows:

### Question-wise Distribution

- **Hard Question**: 6 Marks
- **Medium Question**: 4 Marks

| # | Question Type | Component | Marks | Total Marks |
|---|---|---|---|---|
| 1 | Question 1: Hard | Code understanding and explanation | 3 | 6 |
| | | Design and Analysis | 2 | |
| | | Results | 1 | |
| 2 | Question 2: Medium | Code understanding and explanation | 2 | 4 |
| | | Results | 2 | |
| | | | Total | 10 |

---

## 5. Submission Guidelines

- Code must be written and tested using **RARS** only.
- Output must match the expected results.
- Plagiarism will result in strict academic penalties.

---

## 6. Learning Outcomes

By completing this assignment, students will be able to:

- Understand RISC-V instruction execution
- Use RARS for assembly-level simulation
- Optimize assembly code for efficiency
- Debug and analyze low level programs

---

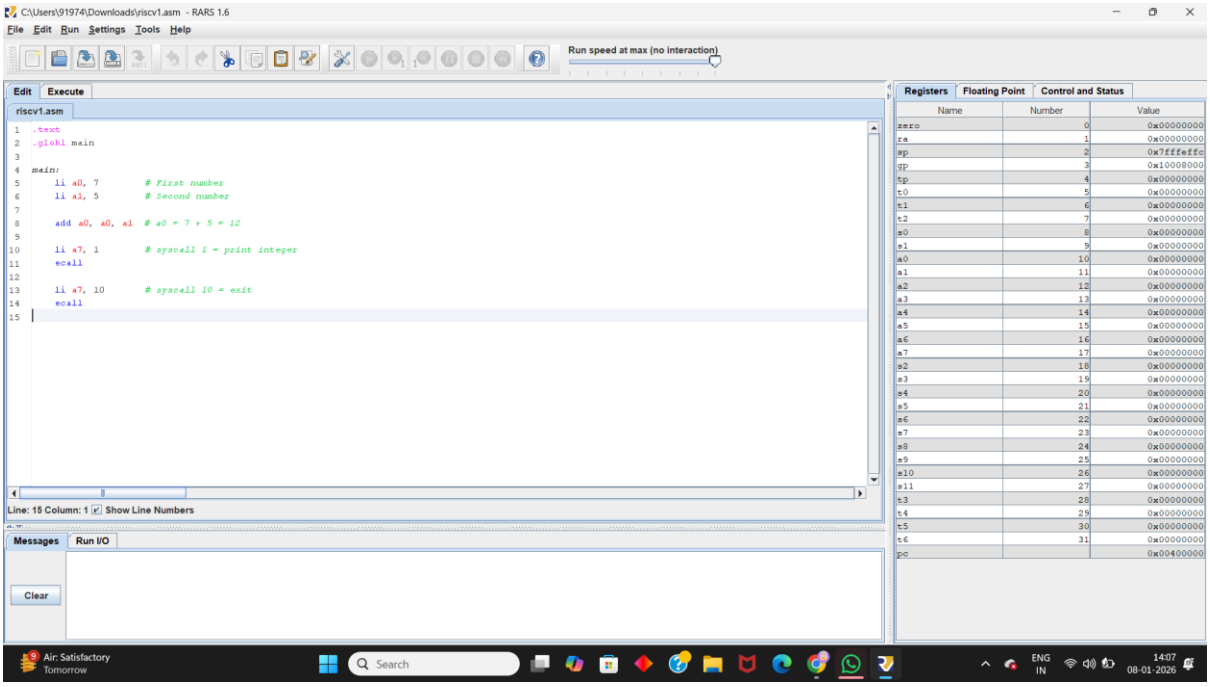
## 7. Conclusion

The Orange Problem Assignment bridges theoretical concepts of processor architecture with practical implementation. Using RARS as a simulation tool allows students to gain hands-on experience with RISC-V, strengthening their understanding of microprocessor fundamentals.

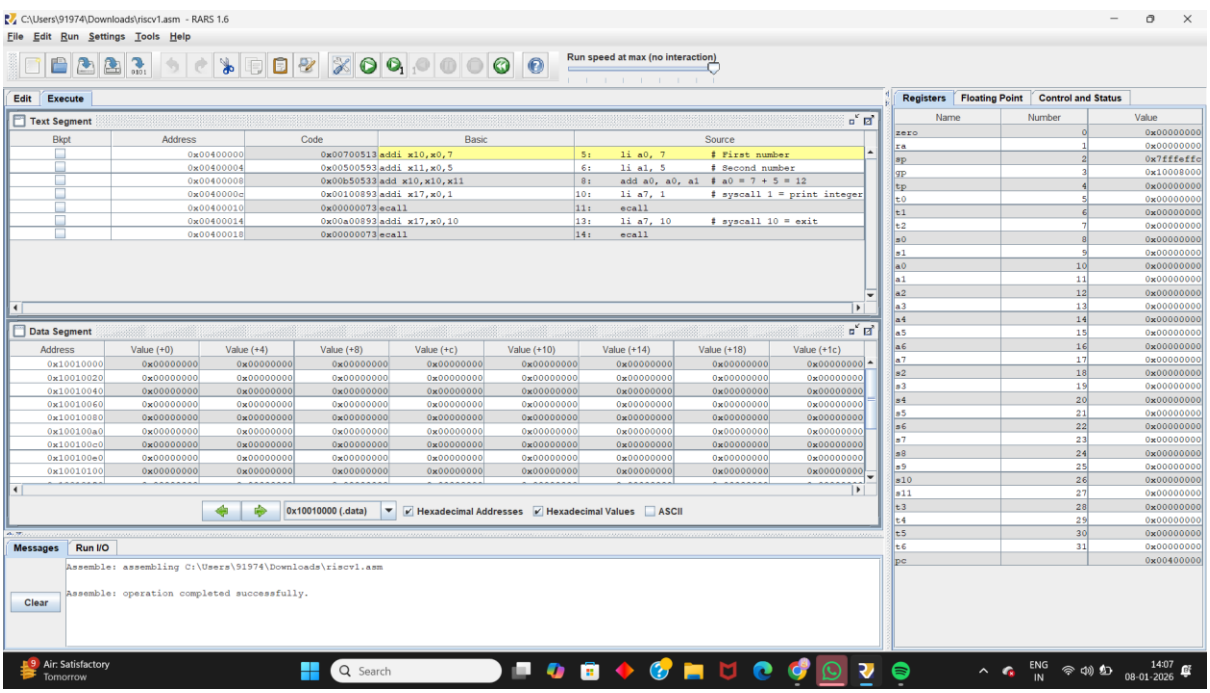
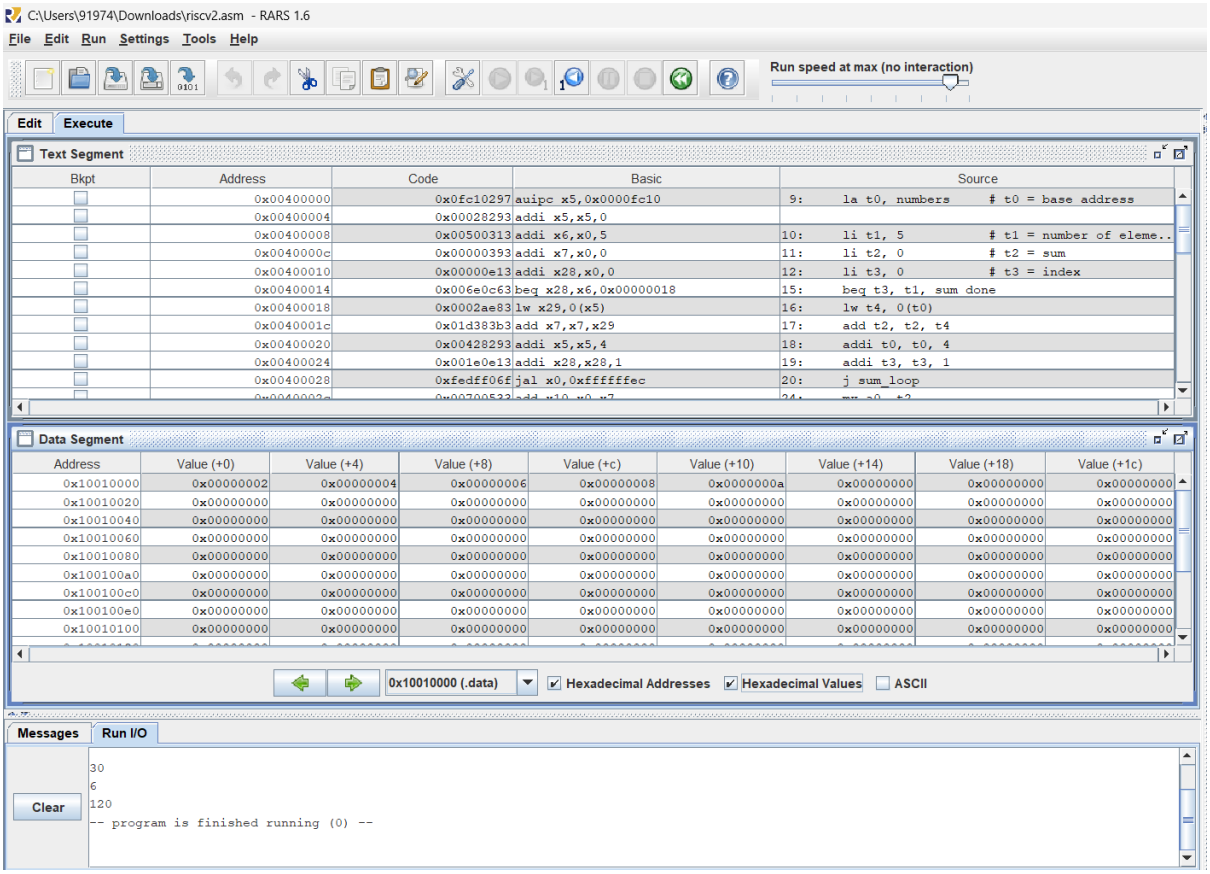
---

## SCREENSHOTS OF RARS SOFTWARE:

Edit Area:




Memory View:



Register view after execution: